

AI and Generative AI Fundamentals

From AI, Machine Learning and Deep Learning to Generative AI — with
enterprise banking use cases

FOR BUSINESS ANALYSTS & TECHNICAL PROFESSIONALS

Large Bank Training Program

Concepts • Architecture • Use Cases • Risk Awareness



Course roadmap

A common language for business and technology teams

01

AI, ML & DL

What each term means and how they relate

02

Generative AI

Foundation models, LLMs and the capability shift

03

Banking use cases

Where AI and GenAI create value across the bank

04

From idea to production

Choosing the right approach, controls and success measures

Why AI matters in banking

AI is not one use case: it is a capability layer across decisions, interactions and operations



Customer interactions

More responsive service, personalization and assisted selling

24/7 +
contextual



Risk & controls

Pattern detection, anomaly identification and decision support

Earlier signals



Operations

Document processing, workflow acceleration and exception handling

Lower friction



Knowledge work

Search, summarization, drafting and analysis across internal information

Faster insight

SECTION 1

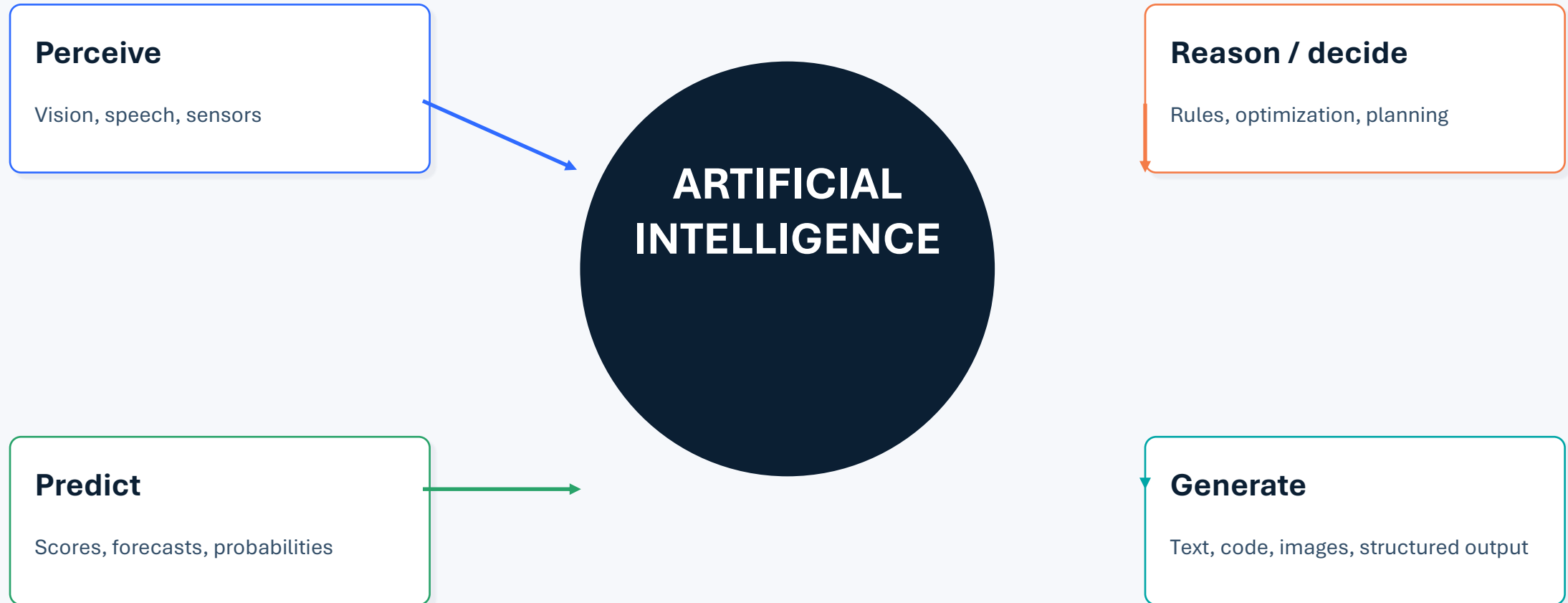
AI, Machine Learning and Deep Learning

Understand the hierarchy before discussing Generative AI



Artificial Intelligence: the umbrella concept

AI is the broad goal of building systems that perform tasks associated with human intelligence



Automation, Machine Learning and Generative AI

Three different ways to make software “smart”

RULES

Rule-based automation

IF / THEN logic

How it works

Humans define the rules

Best suited for

Stable, explicit processes

Banking example

Payment routing rules

PREDICT

Machine Learning

Predict from patterns

How it works

Data teaches the model

Best suited for

Classification, scoring, forecasting

Banking example

Credit risk score

GENERATE

Generative AI

Create / transform content

How it works

Large pretrained models + context

Best suited for

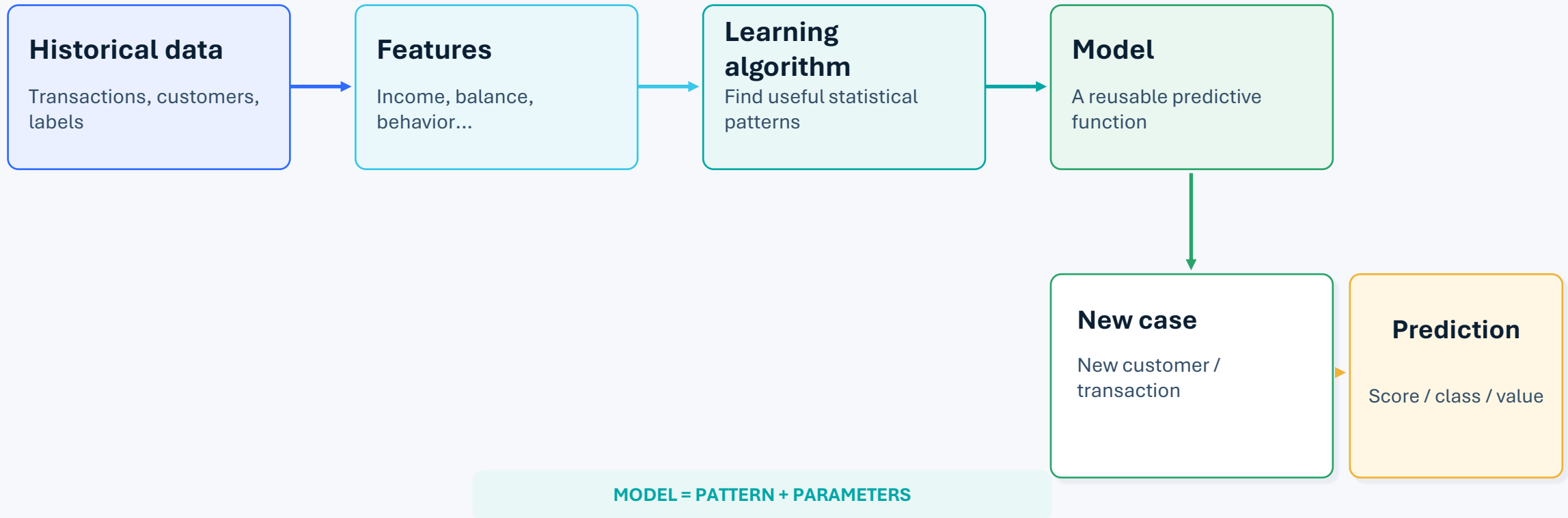
Language, documents, code, multimodal

Banking example

Draft a client call summary

Machine Learning: learning patterns from data

Instead of coding every decision rule, we train a model from examples



Common Machine Learning task families

The target business question determines the learning problem



Classification

Which class?

Banking example

Fraud / not fraud

Output

Binary or multi-class output



Regression

How much?

Banking example

Expected loss amount

Output

Continuous numeric output



Clustering

Which natural group?

Banking example

Customer segments

Output

No predefined label



Anomaly detection

What looks unusual?

Banking example

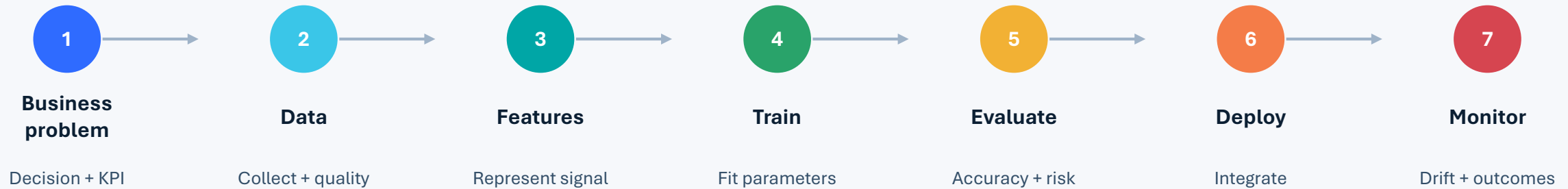
Suspicious transaction pattern

Output

Rare / atypical cases

The Machine Learning lifecycle

Model performance is only one part of a production system

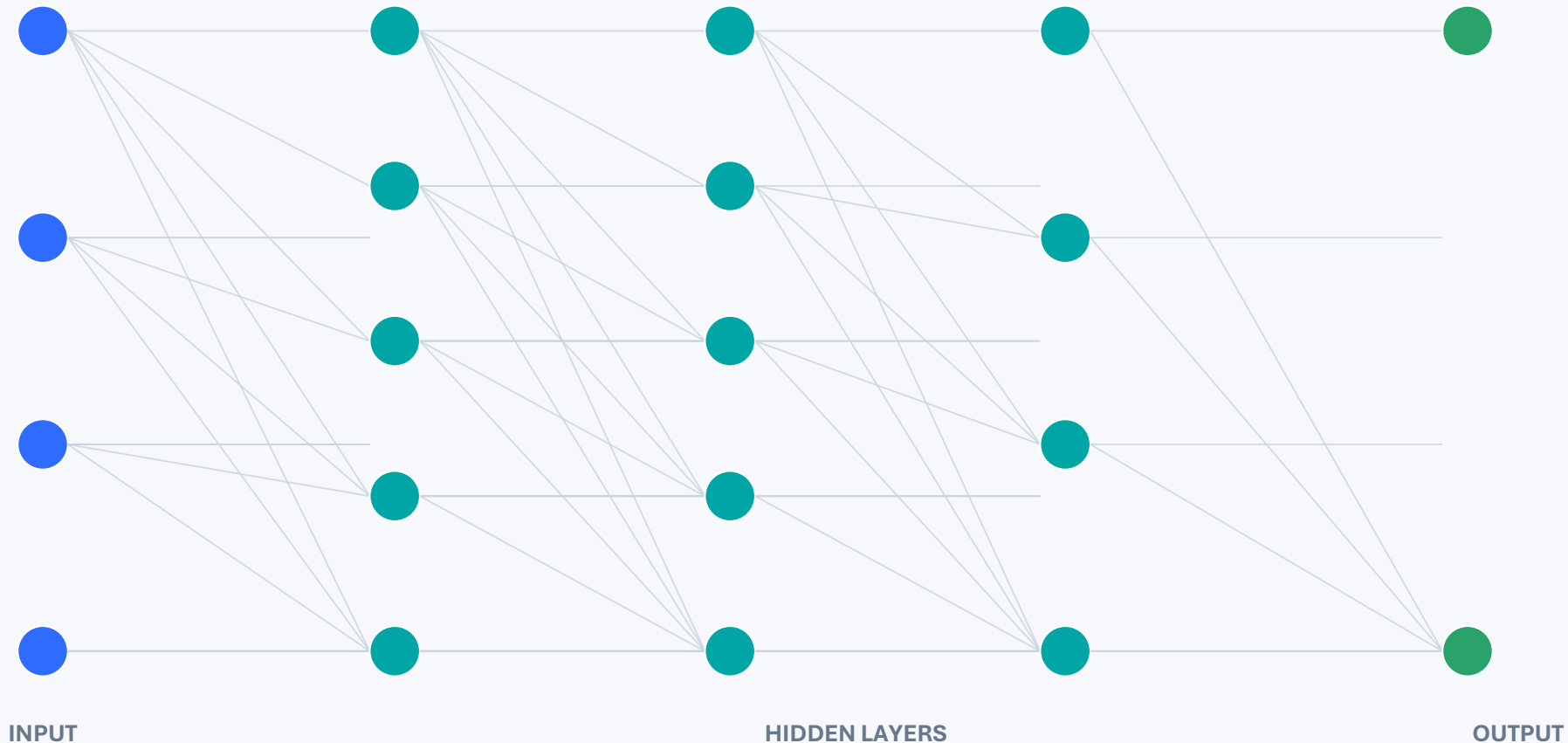


Banking reality:

Data lineage, explainability, validation, controls, access management and monitoring are part of the solution — not afterthoughts.

Deep Learning: learning representations automatically

Neural networks reduce the need for manually engineered features, especially for complex data

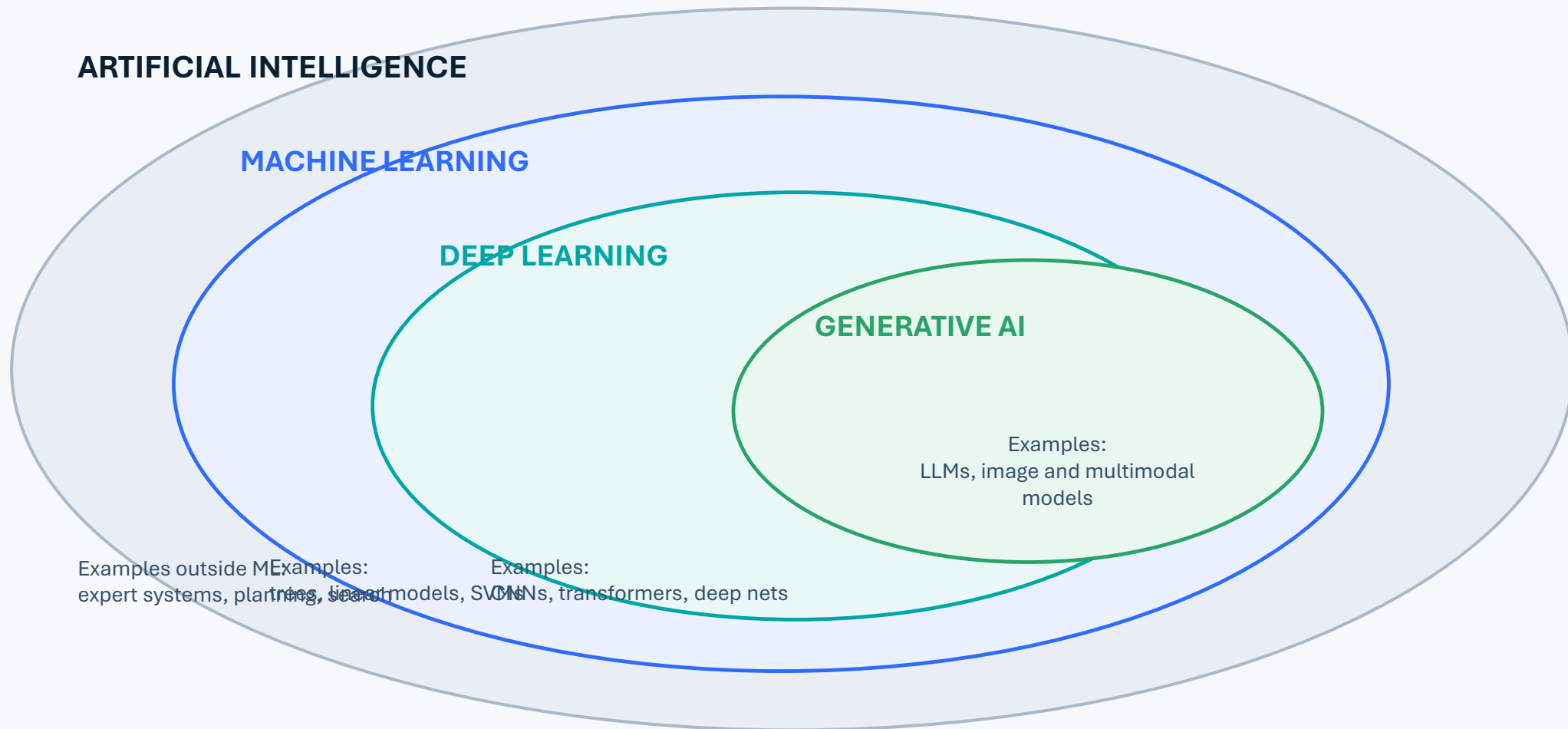


Why “deep”?

- Many stacked layers
- Learns increasingly abstract features
- Scales well with large data and compute

How AI, ML, Deep Learning and GenAI relate

Generative AI overlaps strongly with Deep Learning, but “generative” describes what a model does



SECTION 2

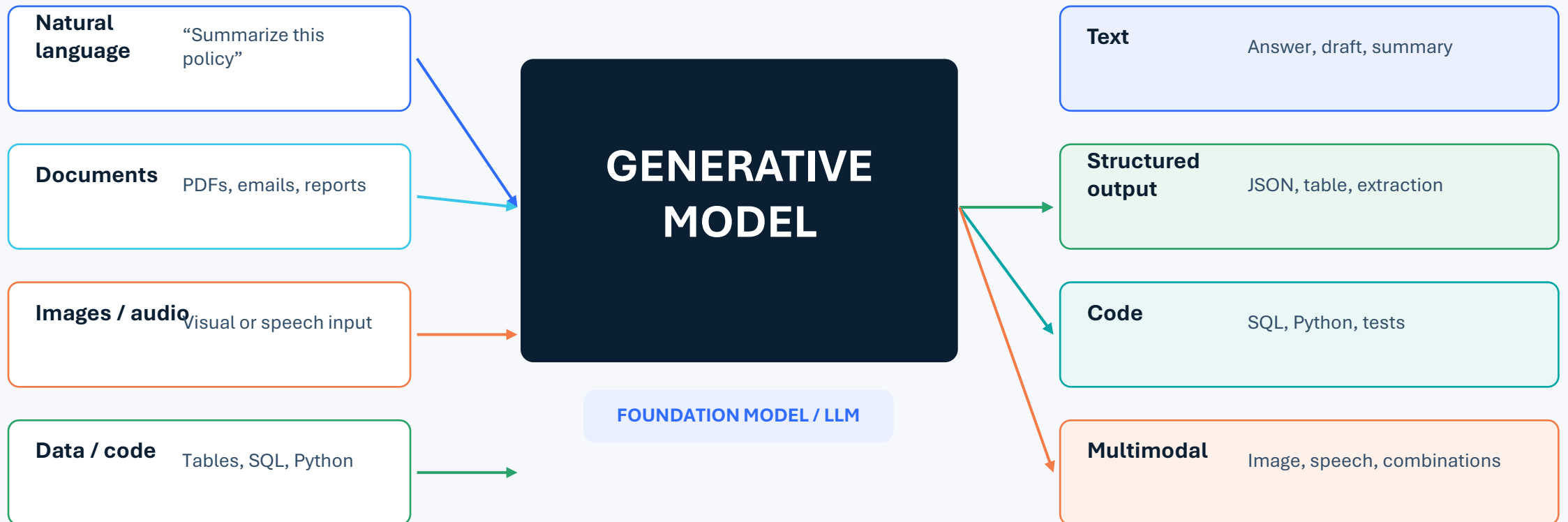
Generative AI and Foundation Models

What changed — and why the change matters for enterprise work



Generative AI: models that create new content

The model produces an output that is statistically consistent with patterns learned during training



Discriminative vs Generative models

Traditional predictive ML estimates an answer; generative models construct a new output

DISCRIMINATIVE

Predict a label or value

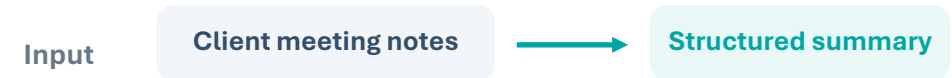


Typical banking tasks

- Credit scoring
- Fraud detection
- Churn propensity
- Forecasting

GENERATIVE

Create or transform content



Typical banking tasks

- Summarization
- Drafting and rewriting
- Extraction from documents
- Conversational assistance

From task-specific models to foundation models

The economic shift: reuse one broad pretrained model across many downstream tasks

BEFORE

Fraud model

Churn model

Text classifier

Document extractor

Separate data, features, training and maintenance for each model

NOW

**FOUNDATION
MODEL**

Pretrained on broad data

Summarize

Extract

Classify

Draft

Q&A

Adapt with prompts, context, retrieval, tools or fine-tuning

What changed with Generative AI?

Five shifts explain why GenAI feels different from previous waves of enterprise AI

1

Natural language becomes an interface Users can ask, instruct and refine rather than learn a specialized UI.

2

One model, many tasks A single foundation model can summarize, classify, extract, draft and reason over context.

3

Zero-shot and few-shot behavior Useful outputs are possible without training a new model for every task.

4

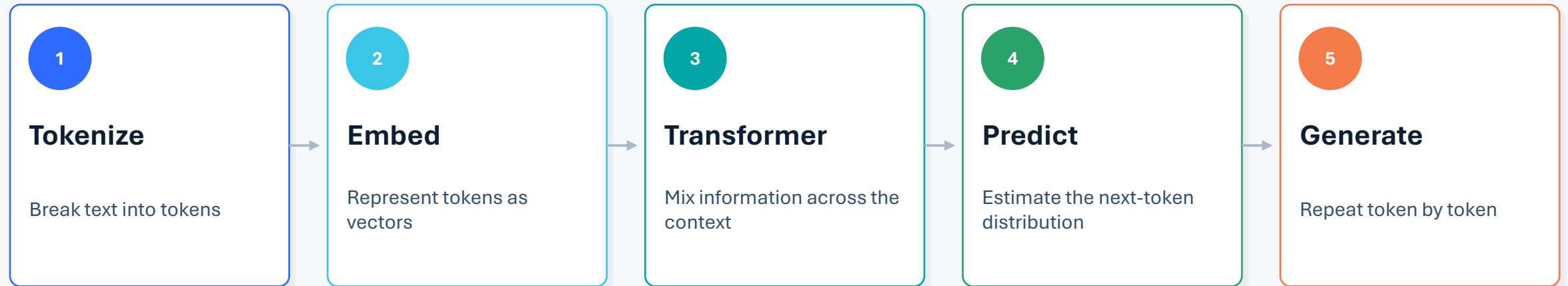
Unstructured data becomes more accessible Documents, emails, policies and conversations become easier to search and transform.

5

AI can use tools and enterprise context Models can retrieve approved information, call APIs and orchestrate workflows.

How a Large Language Model works — high level

The core training objective is simple; the learned behavior is powerful

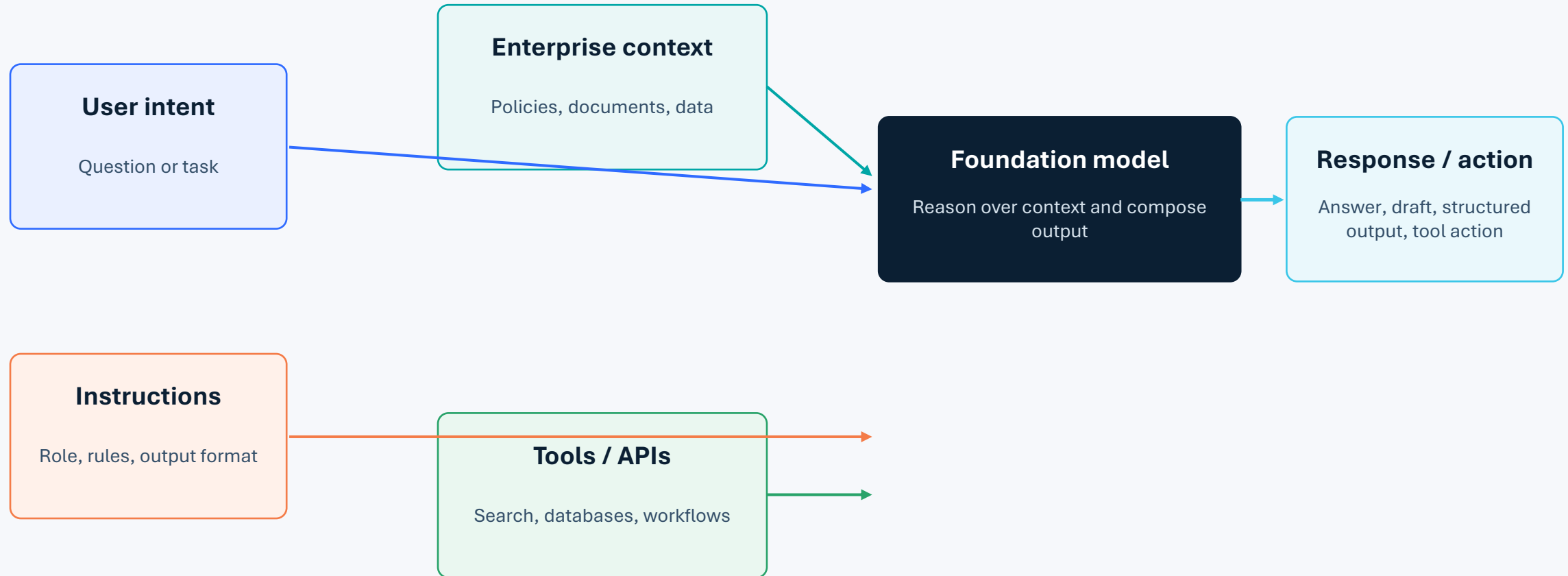


Key idea

LLMs do not retrieve a stored “answer” from a database by default. They generate the most plausible continuation given the model and the context.

Prompt, context, grounding and tools

Enterprise value comes from combining the model with controlled context and systems



Prompting vs RAG vs Fine-tuning

Three different adaptation mechanisms — often used together

Approach	What changes?	Strength	Typical need	Banking fit
Prompting	Tell the model what to do	Fastest; no training	Good instructions, formatting, examples	Most tasks start here
RAG	Retrieve trusted enterprise information into context	Keeps answers grounded and current	Knowledge bases, policies, product documentation	Enterprise Q&A, policy assistance
Fine-tuning	Adjust model parameters using curated examples	Changes style or task behavior	Consistent formats, specialized classification, domain behavior	When prompting alone is not enough

Generative AI: capabilities and limitations

The same flexibility that makes GenAI useful also creates new control requirements

STRONG AT

- Summarizing and transforming language
- Drafting and rewriting
- Extracting information into structured formats
- Explaining, brainstorming and Q&A
- Working across large amounts of unstructured content
- Generating code or queries with human review

WATCH FOR

- Hallucinations and unsupported claims
- Prompt sensitivity and inconsistent output
- Confidentiality and data leakage risks
- Bias and uneven performance across cases
- Over-reliance by users
- Need for permissions, logging and human accountability

SECTION 3

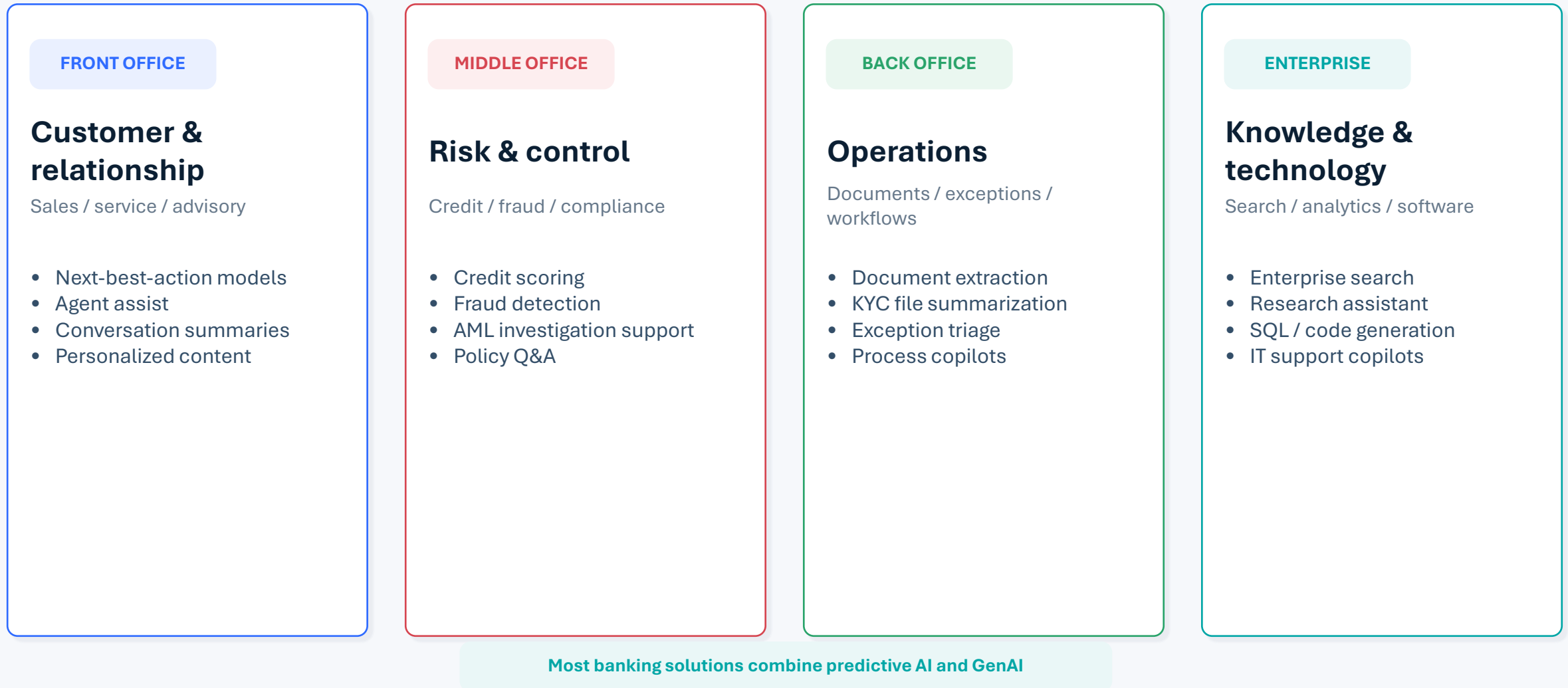
Enterprise Banking Use Cases

Match the AI technique to the business problem — not the other way around



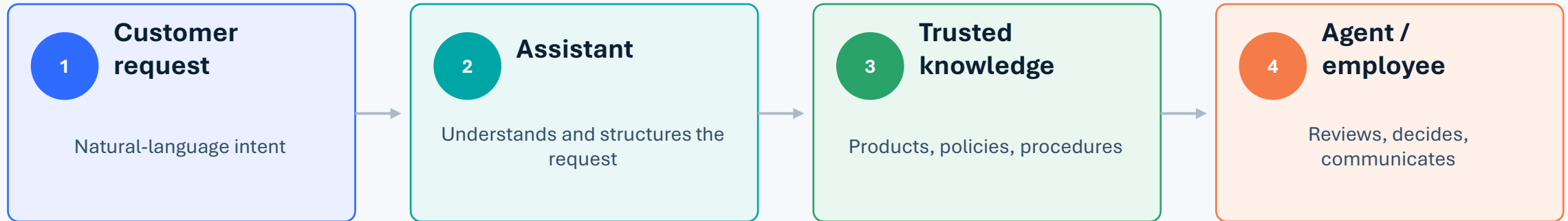
Banking AI use-case landscape

Predictive AI and Generative AI are complementary across the bank



Use case pattern: customer and employee assistance

GenAI is most valuable when it augments people with context, not when it replaces accountability



Typical banking examples

Call-center agent assist

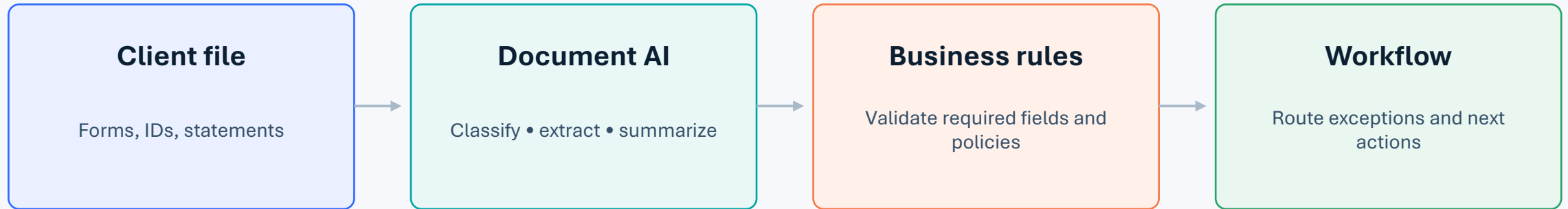
Relationship-manager briefing

Product / policy Q&A

Meeting and call summaries

Use case pattern: document and operations intelligence

Banking processes contain large volumes of semi-structured and unstructured information



High-value examples

- | | | | | | |
|----------|-------------------------------|---|----------|---------------------------------------|---|
| 1 | KYC / onboarding files | Extract data, identify missing evidence, summarize exceptions | 2 | Loan documentation | Summarize covenants, compare terms, extract key clauses |
| 3 | Operations exceptions | Triage free-text cases and propose next steps | 4 | Regulatory / internal policies | Find relevant passages and explain procedures |

Use case pattern: risk, compliance and fraud

Use predictive models for signals and GenAI for investigation support, evidence synthesis and communication

PREDICTIVE / ANALYTICAL AI

Detect and prioritize

- Fraud probability scoring
- Credit risk estimation
- Transaction anomaly detection
- Customer risk segmentation
- Alert prioritization

GENERATIVE AI

Investigate and explain

- Summarize alert evidence
- Draft case narratives for review
- Search policies and procedures
- Compare documents and explanations
- Assist analysts with structured investigation checklists

Human decision authority remains explicit

Use case pattern: technology and analytics productivity

Internal copilots can compress cycle time — provided outputs remain reviewable

DATA

Data analysis

Generate SQL, explain queries, summarize findings

Primary users: Analysts

CODE

Software engineering

Code generation, tests, documentation, refactoring

Primary users: Developers

OPS

IT operations

Incident summaries, runbook assistance, ticket triage

Primary users: Operations teams

KNOW

Enterprise knowledge

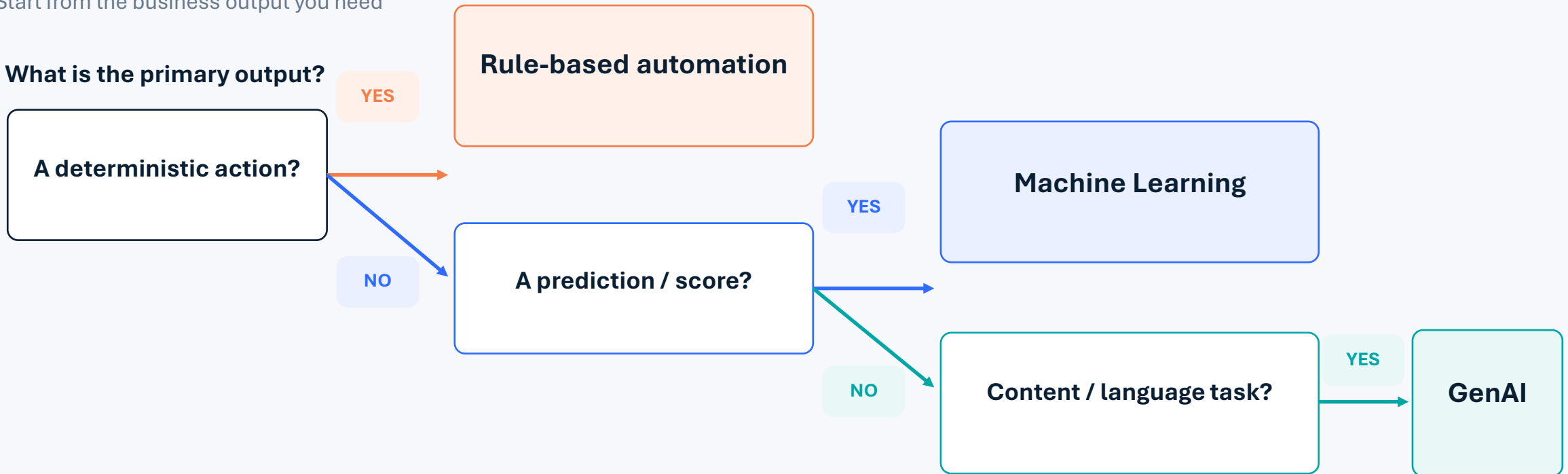
Search, compare and synthesize internal information

Primary users: All knowledge workers

Choosing the right AI approach

Start from the business output you need

What is the primary output?



Most enterprise solutions are hybrid: rules + ML signals + GenAI interface + retrieval / tools + human controls.

A practical enterprise GenAI architecture

The model is one component inside a governed banking solution

User channels

Employees • customer channels • apps

AI application layer

Prompting • orchestration • workflow • agents

Grounding & tools

RAG • search • APIs • databases • calculations

Model layer

Foundation models • predictive ML models

Enterprise data

Policies • documents • transactions • metadata

Controls across all layers

Identity • permissions • logging • evaluation • human oversight • monitoring

Architecture principle: separate business logic, enterprise knowledge, model choice and controls so each can evolve independently.

From PoC to production: five questions to answer

A useful demo is not yet an enterprise capability

1

Value

What measurable business outcome improves?

2

Fit

Is GenAI actually better than rules or predictive ML?

3

Data

What context can the solution access, and under which permissions?

4

Risk

What can go wrong, and where is human review required?

5

Operations

How will we evaluate, monitor and update the system?

KEY TAKEAWAY

AI is a portfolio of techniques. Generative AI expands what can be automated and assisted — especially around language and knowledge — but value comes from the full system around the model.